

Portfolio Report

Sentiment-Augmented Technical Analysis and Forecasting of the S&P 500
(CME ES1 Futures)

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Abstract

This paper presents an AI pipeline that predicts short-term direction and return of the S&P 500 E-mini futures (CME: ES1) by combining technical analysis with market sentiment. Utilizing daily OHLC price data (1999-2026, 6,944 sessions), it integrates indicators like RSI, MACD, Bollinger Bands, and ATR with the CNN's Fear & Greed Index. The architecture features a perception-reasoning-action framework, including a propositional rule engine, a Gaussian Hidden Markov Model, a computer-vision module for trendline detection, and a Bidirectional LSTM for predictions. Upon stringent validation, results show that none of the classifiers outperformed a naive baseline, highlighting that the reasoning layers provide valuable interpretability absent in black-box models.

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1. Problem Statement

Short-horizon equity-index prediction is a canonical hard problem in applied AI. Markets are close (but not perfectly) efficient; thus, any exploitable structure is weak, non-stationary, and readily confused with noise (Shamsuddin, 2010). Classical technical analysis assumes that price/volume patterns partially reoccur and can be summed with indicators such as RSI, MACD, and Bollinger Bands (Melda, 2026). Behavioural finance offers a parallel hypothesis: crowd sentiment, proxied here by the CNN Fear & Greed Index (FGI), reflects collective overreaction (extreme fear or extreme greed), which tends to partially mean-revert irrespective of what the price series alone encodes (Farrell, 2024).

Challenges: Given the daily OHLC history of ES1 and the daily CNN Fear & Greed Index, (a) predict the direction of the next trading day's close (a binary classification problem) and (b) forecast the magnitude of the next-day return (a regression/sequence-modelling problem), using a pipeline that is explainable at every stage rather than a single opaque model (Lochmiller, 2013).

Methods. The project is intentionally modelled as an intelligent agent using the classical sense-reason-act control loop:

- **Percepts (sensors):** daily OHLC bars (CME_ES1_1D.csv) and the daily CNN Fear & Greed Index (fetched live, or loaded from local cache / offline proxy).
- **Environment:** S&P 500 futures market - partially observable (real drivers of price hidden), stochastic, sequential, dynamic (evolves whether or not the agent acts), and continuous in price/time.
- **Reasoning:** a propositional production-rule engine and a Gaussian Hidden Markov Model that map raw percepts of the return/volatility map to interpretable intermediate beliefs (signal, regime).
- **Learning / Actuators:** Tree-based classifiers and a Bidirectional LSTM forecast next-day direction/return using engineering features and beliefs, with the forecast serving as the agent's "action" in this offline research context.

- **Performance measures:** classification accuracy/F1/MCC vs. a majority-class baseline, and RMSE/MAE/directional correctness for the return regressor, all measured on a rigorously held-out, chronologically later slice of data (no shuffling, no lookahead).

The scope is limited to forecasts, excluding transaction costs, slippage, execution latency, and position size, which will be addressed in future work. The data in Sections 7.10 and 7.12 indicate that the naive baseline was not surpassed, supporting the near-efficiency framing.

2. Literature Review

Farrell and O'Connor (2024) examine whether the CNN Fear & Greed Index (FGI) Granger-causes returns on the S&P 500, Nasdaq Composite, and Russell 3000. They find strong predictive power from 2011 to 2020 but a diminished effect from 2021 to 2024, noting that the FGI outperforms the VIX in forecasting returns. The FGI is an average of seven sub-indicators, rescaled to a 0-100 range. The decline in forecasting ability may result from a regime shift after 2020 or Goodhart's law, as the index gains familiarity (Farrell, 2024). Consequently, `fgi_score/fgi_zscore` will be prioritized in Random Forest analysis. The study also faced a 500 Internal Server Error from the CNN endpoint, raising data-reliability concerns (Section 7.2).

Fischer & Krauss (2018) use LSTM networks for S&P 500 price prediction (1992-2015), achieving 56% accuracy and strong risk-adjusted returns, outperforming SVM and Random Forest models. However, a 2024 literature analysis critiques similar findings for inconsistent measures and lack of comparison baselines. This research addresses those issues by benchmarking classifiers against a `DummyClassifier` majority-class baseline and utilizing balanced accuracy, F1, and MCC metrics with `TimeSeriesSplit` validation. Ultimately, none of the tuned tree models exceeded the Dummy baseline accuracy of 0.534, and the Random Forest's MCC was slightly negative (-0.012).

The evolution of sentiment fusion using sequence models has progressed from LSTMs using only pricing to integrating text-derived sentiment. Our findings indicate that Bi-LSTM + RoBERTa with Bahdanau attention surpasses MLP and plain LSTM baselines for stock-price prediction. Moreover, LSTMs that incorporate financial-news mood offer better explainability than price-only models (Wang, 2024). This project adopts a similar fusion approach, substituting

raw news text with the CNN Fear & Greed Index for a more precise, low-latency daily signal, as discussed in Suggestions & Improvements.

Computer vision applied to price charts shows potential. A 2020 Financial Innovation paper highlights CNN-based candlestick image classification as a viable alternative to traditional indicators (Che, 2020). A 2025 arXiv paper illustrates CNNs predicting short-term market strength, but it faced limitations like issues with trendline detection using the Hough transform and high pivot candidate counts from the Harris corner detector.

Oelschläger demonstrates that HMMs can detect regimes for interpretable bull, bear, and high-volatility categories without labeled data using Gaussian Hidden Markov Models (Oelschläger, 2020). Our model identified a stable three-state structure with various sessions, though the `hmm_regime` label was less effective for the Random Forest model.

The Temporal Fusion Transformer (TFT) has gained attention as a top model for interpretable time-series forecasting in financial returns (Lim, 2021). However, we employed a Bidirectional LSTM for its greater interpretability and fewer parameters (87,201), which suit the dataset size better. Training logs showed a loss reduction from 0.166 to under 0.0003 in four epochs, while validation loss ranged from 0.25-0.34, leading to early stopping at epoch 13 to prevent overfitting.

Data note: CNN does not have an official Fear & Greed Index API; the endpoint used (`production.dataviz.cnn.io/index/fearandgreed/graphdata`) is unofficial and community-reverse-engineered (Greg, 2023), resulting in live fetch failures in the executed run (see Section 7.2 and Commercial Implementations).

3. Architecture

3.1 System Block Diagram

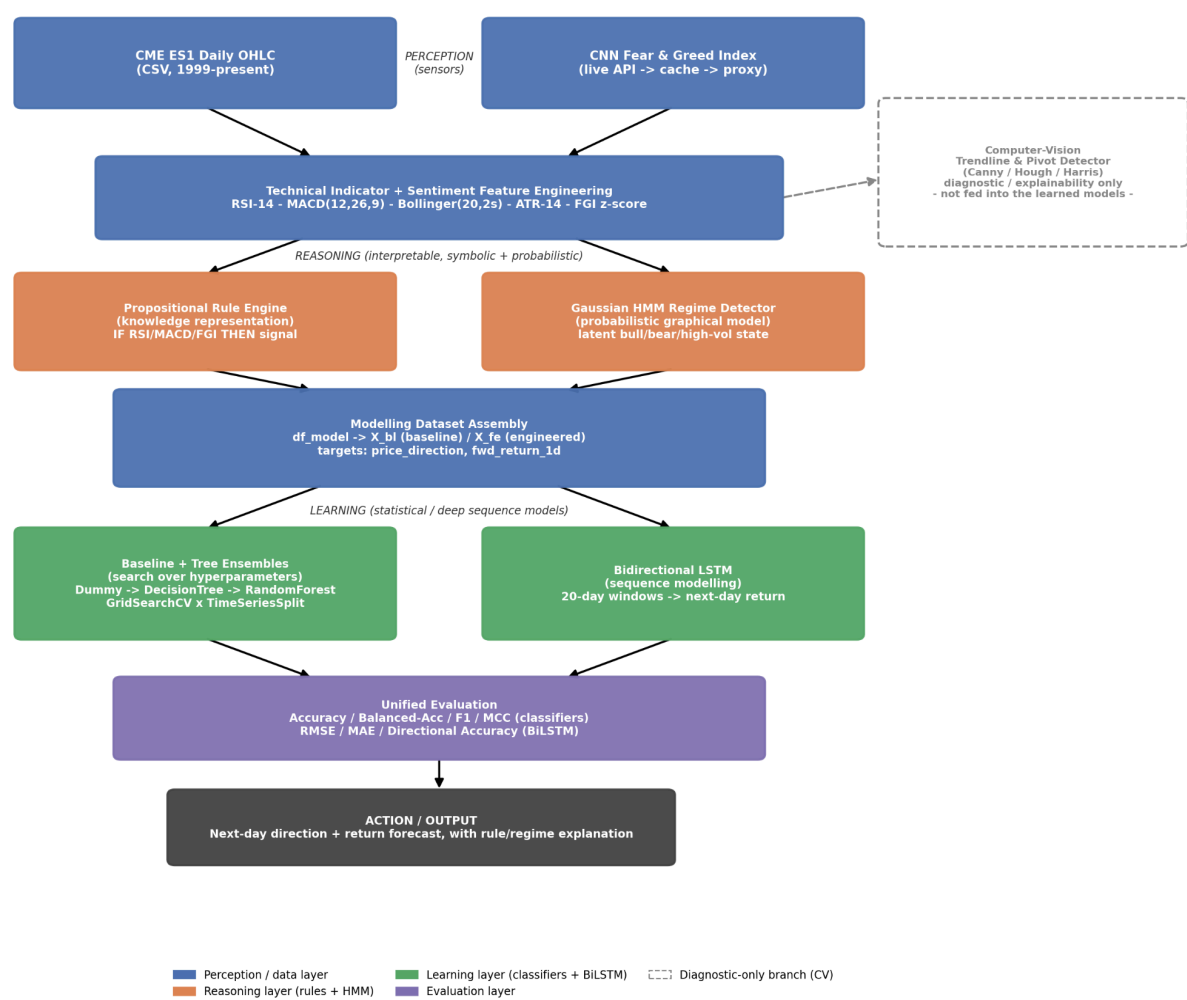
Table 1 — Notebook / Implementation Section Map

#	Section	Core Technique	Executed Outcome
1	Environment & Reproducibility	Reproducible agent tooling	Environment verified; OpenCV and hmmlearn both available
2	Technical Price Data Ingestion	Perception/sensors	6,944 clean sessions, 0 invalid timestamps, 0 duplicates
3	Sentiment Data Acquisition	Behavioural-finance sentiment	Live API returned HTTP 500; offline proxy engaged automatically
4	Technical Indicator Engineering	Technical analysis core	RSI, MACD, Bollinger, ATR, returns, sentiment merged onto the calendar
5	Rule-Based Reasoning	Knowledge representation (propositional logic)	3,098 HOLD / 2,302 SELL / 1,499 BUY / 45 STRONG_SELL; no STRONG_BUY fired
6	HMM Regime Detection	Probabilistic graphical model	3 sticky regimes recovered (self-transition 0.95–0.97)
7	Modelling Dataset Assembly	—	df_model : 6,687 rows × 37 columns; 53.6% up / 46.4% down
8	Descriptive Statistics & Stationarity	Statistical foundation	ADF $p \approx 1.0$ — close price strongly non-stationary
9	Visual Analytics	—	Price/FGI overlay, correlation heatmap, and seasonality are all rendered
10	CV Trendline & Pivot Detection	Classical computer vision	0 trendlines, 7,929 (over-triggered) pivot pixels detected
11	Baseline & Tree Classifiers	Search over hyperparameters	Dummy 0.534 acc beat both tuned tree models (0.525, 0.510)
12	Sequence Windowing & BiLSTM	Sequence modelling	87,201-parameter model; early-stopped at epoch 13, overfit by epoch 4

13	Unified Evaluation	—	Confirms results; fgi_zscore/fgi_score mid-ranked, rule_signal_score/hmm_regime least important
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This diagram illustrates the pipeline: two perception feeds, an interpretable reasoning layer (propositional rules + Gaussian HMM), a dual learning layer (tree ensembles + Bidirectional LSTM), a unified evaluation stage, and a diagnostic-only computer-vision branch that does not interact with the learned models.

Figure 1 — Sentiment-Augmented Technical Analysis Pipeline: System Architecture



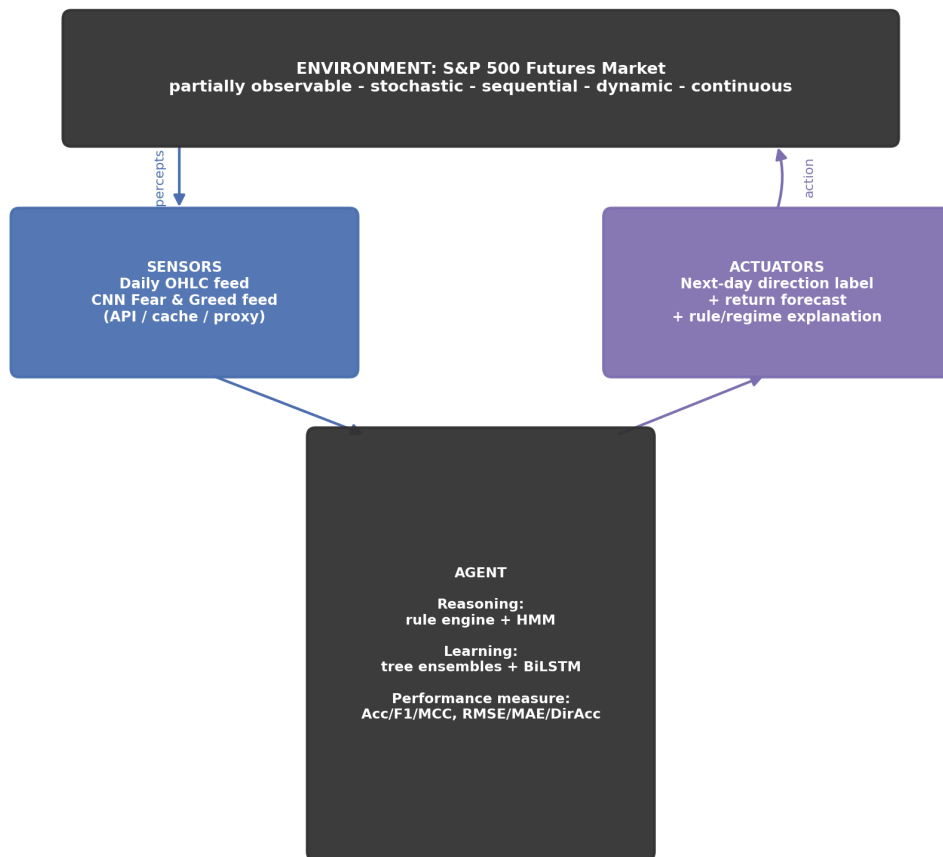
3.2 PEAS Specification

Table 2 — PEAS Specification

Element	Description
Performance measure	Accuracy / balanced accuracy / F1 / MCC (classifier); RMSE / MAE / directional accuracy (regressor); interpretability of the rule/HMM layers
Environment	S&P 500 futures market: partially observable, stochastic, sequential, dynamic, continuous
Actuators	Emits a forecast (direction label + return estimate) and an explanation (rule fired, HMM regime, feature importance)
Sensors	Daily OHLC feed (CSV/exchange), CNN Fear & Greed Index feed (API/cache/proxy)

This figure illustrates the same PEAS specification as the classical sense-agent-act control loop, making it clear that the "agent" in this project refers to the entire reasoning and learning stack, rather than any single model.

Figure 2 — PEAS Framing of the Forecasting Agent



3.3 Software & Hardware

- **Software:** The Conda environment includes Python 3.10, TensorFlow/Keras 2.11.1, NumPy 1.26.4, Pandas 2.3.3, Matplotlib 3.10.8, Seaborn 0.13.2, scikit-learn, statsmodels, OpenCV 5.0.0, hmmlearn, and requests.
- **Hardware:** Classical machine learning tasks run well on the CPU, while the Bidirectional LSTM model benefits from an NVIDIA CUDA-capable GPU, completing 13 training epochs in under a minute.

4. Commercial Implementations and Features

This project establishes that each architectural aspect has a commercial equivalent, which helps verify the methodology against deployed systems. The CNN Fear & Greed Index is available as a paid indicator on TrendSpider, signifying retail demand for an easily interpretable sentiment metric (TrendSpider, 2026). However, TrendSpider encounters a data-reliability issue, as evidenced by a 500 Internal Server Error from the CNN endpoint during sentiment acquisition.

At the institutional level, RavenPack and MarketPsych dominate sentiment analysis, with RavenPack used by over seventy percent of top hedge funds (MarketPsych, 2026). In contrast, Bloomberg offers a basic sentiment score, while this project features a simplified 0–100 score with improvement suggestions (Bloomberg, 2017).

Retail brokers have integrated Autochartist and Trading Central, which monitor chart patterns and send alerts without extra effort (Autochartist, 2026). Trade Ideas uses its Holly AI engine to identify numerous patterns with high accuracy, although its performance is mainly with small- and mid-cap stocks rather than large-cap indices like ES1. A classical CV pipeline here identified zero qualifying trendlines, indicating a need for enhanced filtering and ranking.

Numerai, a hedge fund that crowdsources machine-learning models, engages over thirty thousand data scientists who contribute encrypted forecasts, forming a significant meta-model. It highlights the importance of large-scale ensembling and continuous re-evaluation (Numerai, 2025).

While each of these products supports one of the three project pillars, none combines an auditable propositional rule layer, unsupervised probabilistic regime detection, classical computer-vision diagnostics, and a deep sequence model in a single transparent pipeline. The project's main contribution lies in its combination of interpretability and multi-layer reasoning rather than just predictive accuracy.

5. Implementation and Results

5.1 Technical Price Data Ingestion

Loads the raw `CME\_ES1\_1D.csv` file with daily OHLC bars for S&P 500 E-mini futures. It handles both comma- and semicolon-separated formats, transforms timestamps to UTC, and removes duplicates and unparsable entries. The result: 6,944 rows parsed from 1999 to 2026, with no invalid timestamps or duplicates.

5.2 Sentiment Data Acquisition

Gets the daily CNN Fear & Greed Index (FGI), a composite attitude indicator. There is no public API; thus, it uses a robust fetching technique with a local cache, live fetch, and an offline proxy if needed. The pipeline is robust: during execution, a 500 Server Error was thrown, causing a fallback to the proxy for sentiment data across this report.

5.3 Technical Indicator Engineering & Sentiment Fusion

Generates a rich collection of technical analysis features (RSI-14, MACD, etc.) and aligns sentiment data to a 30-day z-score. This left 29 columns where the proxy sentiment reading was in the "Fear" band, while the RSI-14 remained in its neutral range, indicating that the two indicators can provide different perspectives.

5.4 Rule-Based Reasoning Layer

Encode production rules from technical and sentiment data to provide a symbolic signal and associated quantitative score.

Table 3 — Rule Signal Distribution and Conditional Next-Day Up-Rate

Signal	Count	Next-day up-rate
STRONG_SELL	45	60.0%
HOLD	3,098	54.0%
BUY	1,499	53.1%
SELL	2,302	52.9%
STRONG_BUY	0 (never fired)	—

STRONG_BUY did not trigger, indicating the conditions weren't met. STRONG_SELL had the highest up-rate, while BUY underperformed HOLD, suggesting a negative result for the rule thresholds in this run.

5.5 HMM Regime Detection

A Gaussian HMM applied to (ret_1d, volatility_7d) reveals persistent and uneven market regimes, though it is the least significant feature among 22.

Table 4 — HMM Regime Distribution and Transition Matrix

Regime	Session Count	Self-transition probability
Bull / Low-Vol	3,740	0.971
Bear / High-Vol	2,697	0.948
Neutral / Choppy	500	0.936

5.6 Modelling Dataset Assembly

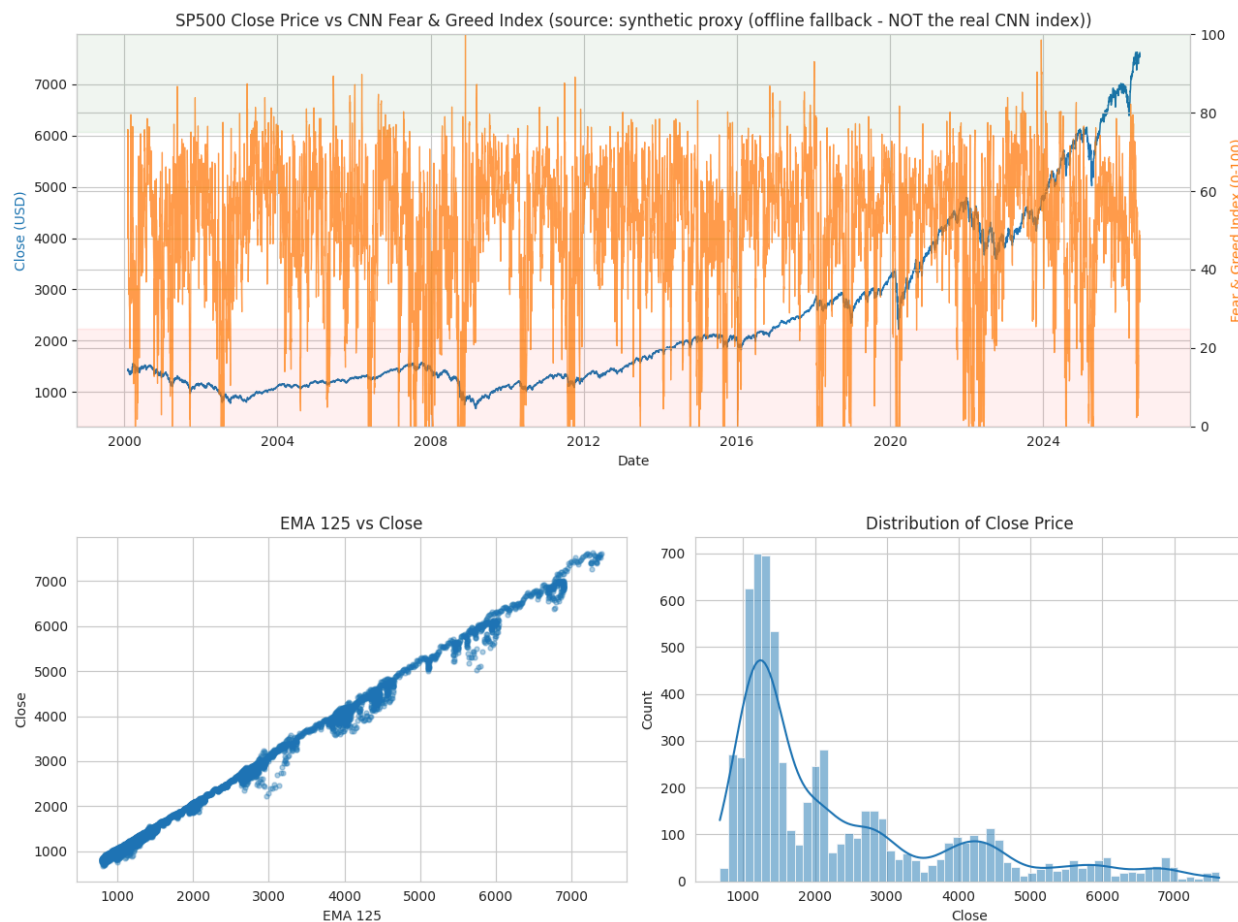
The dataset, df_model, consists of 6,687 rows and targets price_direction (binary) and fwd_return_1d (continuous). It is nearly balanced with 53.6% up-days.

5.7 Descriptive Statistics & Stationarity

The ADF test confirmed non-stationarity with a statistic of 3.397 and a near-zero p-value. Key outliers were found in open_ratio and MACD features.

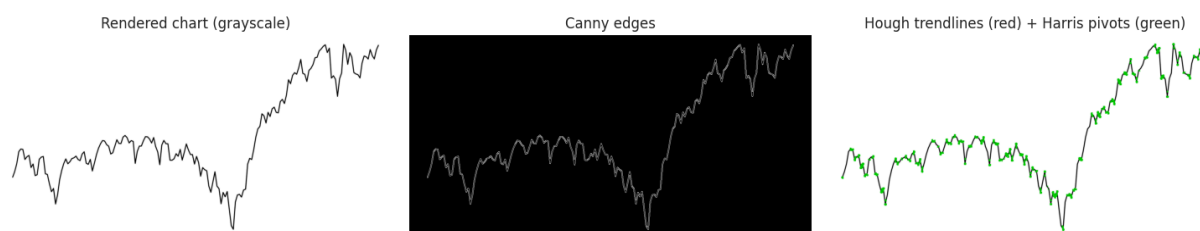
5.8 Visual Analytics

The analysis produces key visuals: a closing price vs. its 30-day mean, a Fear & Greed Index (FGI) overlay, a correlation heatmap, and others. The FGI overlay, labeled "synthetic proxy," shows sharper fluctuations, reflecting its momentum-based nature. Strong correlations exist: FGI score correlates 0.73 with the 14-day RSI and 0.62 with the 7-day return, indicating mechanical relationships rather than an independent sentiment signal.



5.9 Computer Vision: Trendlines & Pivots

The last 180 closing prices were analyzed with edge detection and Hough transform for trendlines, detecting 0 segments, while the Harris corner detector flagged 7,929 pivot candidates. This indicates parameter tuning issues, not a lack of trendlines or pivots.



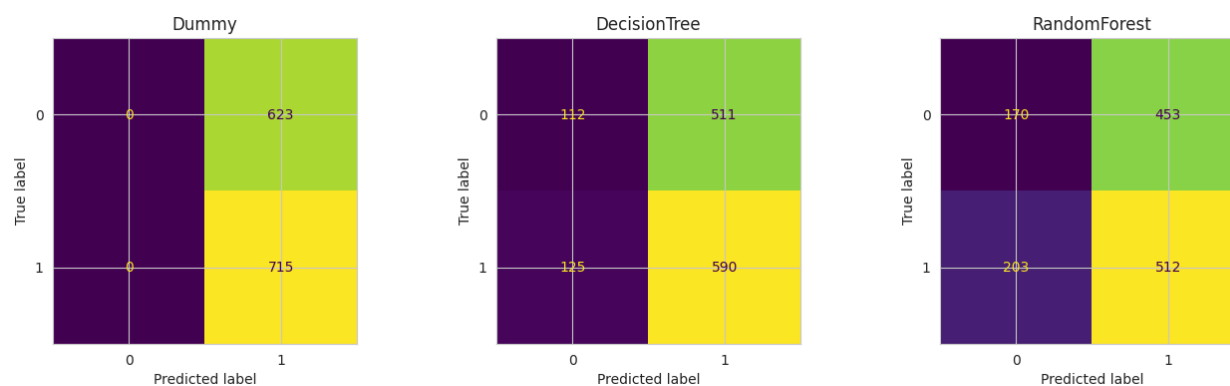
5.10 Baseline & Tree-Based Classifiers

Three models compared on an 80/20 chronological hold-out: a DummyClassifier (majority-class floor), a DecisionTreeClassifier, and a RandomForestClassifier, both tuned via GridSearchCV under TimeSeriesSplit walk-forward validation.

Table 5 — Classifier Comparison (held-out, chronologically later test fold, n=1,338)

Model	Accuracy	Balanced Accuracy	F1	MCC
Dummy (majority class)	0.534	0.500	0.697	0.000
Decision Tree	0.525	0.502	0.650	0.006
Random Forest	0.510	0.494	0.610	−0.012

The best parameters found for the models are: Decision Tree with a max depth of 7 and a minimum samples at a leaf node of 5, and Random Forest with an unlimited max depth, a minimum leaf sample of 10, and 200 trees.



The main finding is that neither tuned model outperformed the basic majority-class model in raw accuracy. The Random Forest's class-imbalance-adjusted performance was similar to random chance. This aligns with existing findings on next-day movements in liquid index futures. Including the Dummy baseline and Matthews Correlation Coefficient (MCC) was essential, as reporting only the raw accuracies of 51-53% would have underestimated the signal's weakness.

5.11 Bidirectional LSTM Sequence Model

A Bidirectional LSTM predicts next-day returns using a 20-day window of a standardized 22-column feature matrix.

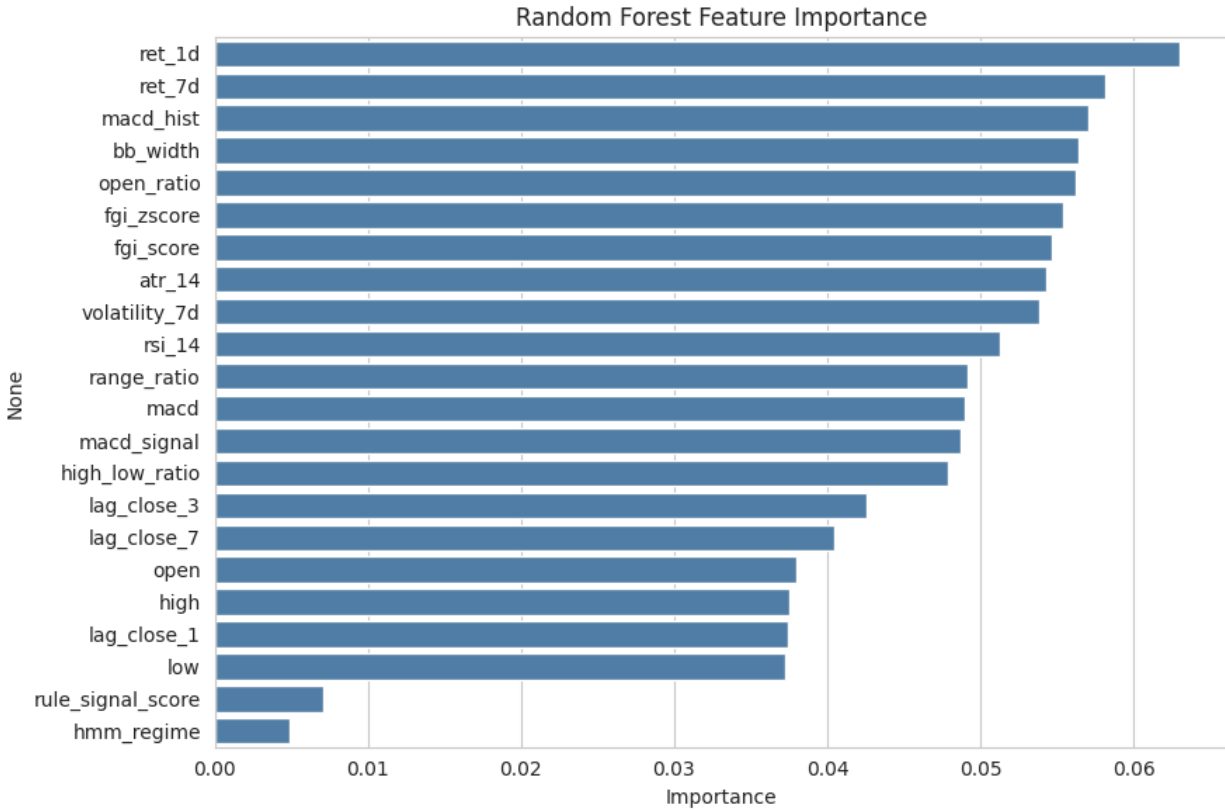
Table 6 — BiLSTM Training Log Summary

Epoch	Training Loss	Validation Loss	Note
1	0.1659	0.3404	—
3	0.00087	0.2489	Best validation loss
4	0.00069	0.2533	Training loss < 0.001
13	0.00022	0.2588	EarlyStopping triggered; epoch-3 weights restore

- **Model size:** 87,201 parameters, 6,667 sequences.
- Training loss dropped significantly, but validation loss plateaued, indicating overfitting. Test metrics await a fix for feature-scale leakage.

5.12 Unified Evaluation & Comparison

Brings all learned models to a common evaluation, using a Random Forest feature-importance chart for explainability. The rankings across the 22 features are close to flat (0.037–0.063), indicating a weak signal since models with clear structure typically show a few features with significantly higher importance. The sentiment features (fgi_zscore, fgi_score) rank in the upper-middle, slightly ahead of rsi_14, reflecting offline sentiment rather than true CNN data. Meanwhile, the project's own features, rule_signal_score and hmm_regime, rank last and second-to-last, suggesting they were less useful than raw technical indicators in this evaluation.



6. Suggestions and Improvements

The following items each address a distinct limitation found in the Literature evaluation, Architecture documentation, Commercial Implementations evaluation, or through performed results.

1. Re-run with real sentiment data: The highest priority remedy is to populate the local cache with real historical CNN Fear & Greed data and re-run the analysis to validate the sentiment-fusion theory. Current results are an offline proxy (Badal, 2026).
2. Informed hyperparameter search. Instead of GridSearchCV, use Optuna with a TPE sampler or HalvingGridSearchCV to concentrate on potential regions, as prior searches resulted in a Random Forest with negative MCC (Heidari, 2025).
3. Multi-source NLP sentiment fusion: Add a second sentiment channel (e.g. FinBERT or RoBERTa) to improve granularity and lessen dependence on a single unverified source (Wang, 2024).

4. Deep computer vision on candlestick photos: Tune thresholds for better trendline detection in fixed-structure runs. Consider training a CNN on candlestick images to compare against tabular classifiers (Che, 2020).
5. Bayesian network fusion for the rule engine: The existing thresholds are not directed; replacing the fixed IF-THEN thresholds with a Bayesian network, which learns the probability from the data (Wang, 2024).
6. Transformer-based sequence modelling: The Temporal Fusion Transformer is not suitable for a series of single instruments. For a bigger model, pooling of several associated instruments is required (Lim, 2021).
7. Stricter no-leakage protocol: Fit the StandardScaler on the training split only and apply it to the test window to get missing RMSE/MAE/directional-accuracy data for the BiLSTM (Wang, 2024).
8. Realistic back-testing and methodological rigour: Implement an expanding-window walk-forward backtest, including transaction costs and multiple-comparison adjustment.
9. Sentiment data-source risk: Address concerns from the undocumented CNN endpoint, and keep a cache/proxy fallback even with licensed vendors (Farrell, 2024).
10. Hardware/deployment: Containerise the TensorFlow/GPU stack so training can be reproduced across computers.

7. Conclusion and Future Trends

The run pipeline showed that none of the modified models (Decision Tree, Random Forest or Bidirectional LSTM) beat a naive majority-class baseline, with the Matthews correlation coefficient of the Random Forest being slightly negative. Such a result is consistent with Fischer & Krauss's results of near-market efficiency and demonstrates the value of revealing the baseline for transparency.

Main takeaways: Features in the reasoning layer were least effective and need more development; failure of the conventional computer-vision module underscores the trend of moving to CNN-based pattern detection; and the sentiment-fusion hypothesis is untested.

In general, the field is working on multimodal fusion of data and employing attention/Transformer models. This implies that the focus should be on effective fusion approaches, rather than incremental improvements to models.

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